Heritage Bank Digital — Technical Context

Platform Architecture

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│                              CLIENT LAYER                                 │
│                                                                           │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐  │
│  │  Mobile App  │  │  Online      │  │  Branch      │  │  SMB Business │  │
│  │ (React Native│  │  Banking     │  │  Teller App  │  │  Portal       │  │
│  │  iOS/Android)│  │  (React SPA) │  │  (React SPA, │  │  (React SPA)  │  │
│  └──────┬───────┘  └──────┬───────┘  │  internal)   │  └──────┬───────┘  │
│         │                 │          └──────┬───────┘         │          │
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          │  HTTPS          │                 │  mTLS (internal) │
          │  + biometric    │                 │                 │
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│            API GATEWAY — Kong Gateway (on-prem + AWS API GW for cloud)    │
│    Okta JWT validation  •  Rate limiting  •  mTLS enforcement  •  Audit log │
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       │            │              │                  │
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│  Customer   │ │  Account    │ │  Payment    │ │  Notification       │
│  Service    │ │  Service    │ │  Service    │ │  Service            │
│             │ │             │ │             │ │                     │
│  Okta +     │ │  Spring Boot │ │  Spring Boot │ │  AWS SES + Twilio + │
│  Spring Boot│ │  → BIH → AS/4│ │  → BIH → AS/4│ │  SNS Push           │
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       │            │             │
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│              BANKING INTEGRATION HUB (BIH)                               │
│        The strangler-fig adapter layer — owns all AS/400 communication   │
│                                                                          │
│  FiServ Open Solutions API (REST)    IBM MQ (async)    SFTP batch jobs   │
│  ← ~60% of operations              ← ACH files        ← statements/reports │
└──────────────────────────────────────────────────────────────────────────┘
                                │
                                │
┌───────────────────────────────▼──────────────────────────────────────────┐
│              LEGACY CORE — IBM AS/400 (IBM i 7.4 / Power9)               │
│                  FiServ Precision Core Banking System                    │
│                                                                          │
│   Accounts (DDA/SAV/CD/HELOC)    Transaction Ledger    Interest Engine   │
│   ACH Module (FedACH)            GL / Regulatory       Loan Module       │
│   Nightly batch (00:00–04:00 CT) EOD cutoff (18:00 CT)                   │
└──────────────────────────────────────────────────────────────────────────┘
                                │
                                │  IBM MQ  +  Direct integration
       ┌────────────────────────┼──────────────────────────────┐
       │                        │                              │
┌──────▼──────┐     ┌──────────▼──────┐          ┌──────────▼──────────┐
│ Card Mgmt   │     │  Fraud &        │          │  External Networks  │
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│ Service     │    │  Compliance Svc   │          │                        │
│             │    │                   │          │  FIS Worldpay (cards)  │
│FIS Worldpay │    │  NICE Actimize +  │          │  EWS / Zelle (P2P)     │
│ token store │    │  Kafka consumer   │          │  TCH RTP (real-time pay)│
│ Spring Boot │    │  + Spring Boot    │          │  FedACH (ACH)          │
└─────────────┘    └───────────────────┘          │  Fedwire (wire)        │
                                                  │  SWIFT (international) │
                                                  │  Fiserv CheckFree (bill)│
                                                  └────────────────────────┘
      │
┌─────▼──────────────────────────────────────────────────────────────────┐
│                    DOMAIN EVENT BUS — Apache Kafka (on-prem Confluent)  │
│   transaction.posted  •  account.opened  •  transfer.initiated          │
│   fraud.alert.raised  •  card.blocked  •  ach.submitted  •  wire.approved │
└────────────────────────────────────────────────────────────────────────┘
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Services Reference

Service	Responsibility	Technology
Banking Integration Hub (BIH)	Central adaptor: translates REST/Kafka calls to FiServ API + IBM MQ; owns AS/400 error mapping	Spring Boot 3.2 (Java 21) + IBM MQ Client
Customer Service	Customer profile, identity verification, KYC status, proxy/joint account relationships	Spring Boot 3.2 + PostgreSQL 15 + Okta
Account Service	Account summary, transaction history, statement retrieval, balance (real-time available vs book)	Spring Boot 3.2 → BIH → AS/400
Payment Service	Internal transfers, ACH initiation, Zelle, RTP, bill pay dispatch, wire initiation	Spring Boot 3.2 → BIH + external networks
Card Management Service	Debit card controls (lock/unlock, travel notice, limits), FIS Worldpay token management	Spring Boot 3.2 + FIS Worldpay REST
Fraud & Compliance Service	Actimize alert consumption, SAR/CTR workflow, velocity rule enforcement	Spring Boot 3.2 + Kafka consumer + Actimize REST
Notification Service	Balance alerts, transaction confirmations, fraud alerts, secure messages	AWS SES (email) + Twilio (SMS) + SNS (push)
Lending Gateway	Consumer loan application intake, status polling from Blend	Spring Boot 3.2 + Blend API
SMB Portal BFF	Backend-for-frontend: multi-account aggregation, user role management, ACH payroll file	Spring Boot 3.2 + GraphQL
Audit Service	Immutable log of all customer-data access and configuration changes (SOX + BSA)	Kafka → Elasticsearch + S3 Glacier

Reporting Service	Regulatory reports (CTR, SAR), Snowflake feed for BI, branch performance	Snowflake Connector + Spring Batch

Tech Stack

Frontend

- **Mobile**: React Native 0.73 (iOS 15+, Android 11+); Expo managed; biometric auth via device secure enclave
- **Online Banking / SMB Portal**: React 18 SPA; Vite 5 build; TypeScript strict; Material UI (MUI) v5 component library
- **Branch Teller App**: React 18 SPA (internal, Chrome-only); no offline requirement; mTLS cert-based auth
- **State management**: Redux Toolkit (online banking — complex multi-account state); React Query v5 (server cache)
- **Forms**: React Hook Form + Yup; multi-step wizard pattern for transfers and loan application

Backend (Modern Layer)

- **Runtime**: Java 21 (Spring Boot 3.2) — chosen for FiServ and IBM MQ SDK availability; virtual threads (Project Loom)
- **API style**: REST (external customer-facing); GraphQL (SMB BFF); async Kafka (inter-service events)
- **Auth**: Okta as IdP; OIDC + PKCE for mobile/web; SAML 2.0 for branch SSO (Active Directory federation)
- **Database**: Amazon RDS PostgreSQL 15 (Customer, Account, Payment, Card services); separate schema per service (not shared DB)
- **Cache**: Redis 7 (ElastiCache) — account summary cache (TTL 30 s, invalidated on transaction event), idempotency keys
- **Event streaming**: Apache Kafka (Confluent Platform, on-prem in primary DC; Confluent Cloud for DR)
- **Queue**: IBM MQ 9.3 — reliable AS/400 message delivery; dead-letter queue monitored by BIH operations team

Legacy Layer (IBM AS/400)

- **Platform**: IBM i 7.4 running on Power9 LPAR; managed by Infrastructure team (not Heritage Digital)
- **Core system**: FiServ Precision — COBOL + RPG programs; no source-level changes by Heritage Digital team
- **Access methods**:
  - FiServ Open Solutions API (REST over HTTPS) — account inquiry, balance, transaction posting (~60% coverage)
  - IBM MQ — ACH file submission, batch job triggers, GL event notifications (async, guaranteed delivery)
  - SFTP batch — nightly statement generation, CTR/SAR report extracts, Snowflake data feed
  - **TN5250 screen scraping**: legacy internal tools only; deprecated; BIH is replacing all screen-scraping integrations by end of programme
- **Batch windows**: nightly (00:00–04:00 CT) and EOD (18:00 CT) are **frozen** — no BIH calls to AS/400 during these windows; services must queue and retry

Infrastructure & DevOps

- **Hybrid cloud**: Core AS/400 and Confluent Kafka on-prem (primary DC: Columbus, OH); modern microservices on AWS (us-east-1)

- **Connectivity**: AWS Direct Connect (10 Gbps) between DC and AWS VPC; BIH deployed on-prem and in AWS (active-active)
- **IaC**: Terraform (AWS resources); IBM i changes via formal CAB process only (no IaC)
- **CI/CD**: GitLab CI — compile → unit test → integration test (Docker Compose with AS/400 stub) → staging → prod (manual gate + CAB ticket)
- **Change management**: SOX ITGC requires CAB approval for all production changes; emergency changes need CISO + CTO sign-off
- **Observability**: Dynatrace APM (full-stack Java tracing); IBM i performance logs via SMF records; PagerDuty on-call
- **Secrets**: HashiCorp Vault (on-prem) — database passwords, IBM MQ credentials, FiServ API keys; AWS Secrets Manager for cloud-only services

Key Architectural Constraints

- **AS/400 batch window**: BIH must **not call FiServ API or IBM MQ** between 00:00–04:00 CT and 18:00–18:30 CT. Payment Service queues operations and replays after window. Clients show "system maintenance" message.
- **Book vs available balance**: AS/400 provides two balances — `bookBalance` (ledger, updated at EOD batch) and `availableBalance` (real-time, via FiServ API). Account Service always shows `availableBalance` to customers; never shows `bookBalance` alone.
- **Idempotency is critical**: duplicate IBM MQ messages and Kafka retries are expected. Every Payment Service operation has an idempotency key (`clientTransactionId`) checked against Redis before forwarding to BIH.
- **No direct AS/400 write from services**: only BIH may write to AS/400. All other services call BIH. This is a hard architectural rule enforced by network ACL.
- **SOX segregation**: developers have no access to production AWS environments or AS/400. All prod changes go through the automated pipeline + CAB approval.
- **Strangler-fig progress**: BIH owns the migration backlog. Current state: Account inquiry, balance, transfers (done). ACH origination (in progress). Wire initiation, CD management (roadmap). IVR and branch teller green-screen (deferred — post Phase 2).

Non-Functional Requirements

Category	Requirement
Availability	99.9% for customer-facing channels (planned maintenance excluded); AS/400 batch windows are known downtime
Latency	p95 balance check < 500 ms; transfer initiation < 1 s; AS/400 calls via BIH < 2 s (FiServ API SLA)
Throughput	1.2 M retail customers; peak 50,000 concurrent sessions (mobile payday surge)
Data residency	All customer data must remain within US (regulatory + contractual constraint)
Audit retention	BSA records: 5 years; SOX ITGC evidence: 7 years; transaction logs: 7 years
RTO / RPO	RTO 4 h (AS/400 failover to DR site); RPO 1 h (MQ journal + PostgreSQL continuous backup)

Batch window	BIH queues all AS/400-bound requests during 00:00–04:00 CT nightly batch and 18:00 CT EOD
Change freeze	No production deployments on Fridays after 15:00 CT or on public holidays (SOX control)